Classification of Bachelor Thesis – supervisor

Author of classification:	prof. Ing. Michal Krátký, Ph.D.
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Opponents:	Ing. Petr Lukáš, Ph.D.
Title:	Trie Data Structure
Thesis version:	1
Student:	Tuan Phong Huynh

1. *Assignment of the thesis.*

   The difficulty of the assignment is average, the solution can include many variants of the implementation, student selected the appropriate ones. It can be a good idea to continue with the work on master thesis. The assignment of the thesis has been completed.

2. *Student's activity during the project completing.*

   The solution has been often consulted, student worked autonomously.

3. *Student's activity during the process of completion.*

   I have read the work many times, however there are still some parts to improve. For example, the terminology is not consistent: in sorted-array node vs Sorted Array, in a trie structure vs in the Trie, and so on. However, the thesis is mostly readable and clear.

4. *Overall evaluation of the thesis*

   Student implemented three implementations of a node including the following data structures: nonsorted array, sorted array, and hash table. Moreover, the Compressed Trie has been also implemented where unique postfixes are stored instead of nodes with one value. The application runs without issues, the preformace seems to be sufficient. There are some unclear points of the implementation, see questions. The source codes in C++ do not include header files, which is not a good idea.

5. *Evaluation of the new findings contribution.*

   This work does not bring novel ideas, but the comparison of various node implementations can be interesting.

6. *Utilization and selection of information sources.*

   The list of references includes appropriate items. Own work is distinguish from the other works. The test on the plagiarism does not detect any important issues.

7. *Summary evaluation.*

   This work is not perfect, it includes 4 implementations of a node in the Trie and a comparison with the sequential array for insert, delete, point query, and prefix query operations. Moreover, student has tried to solve all issues to appear during the work. As a result, I suggest the evaluation 1.

8. *Question for the defense of the thesis.*

   1. Encoding of unicode characters in nodes is not clear. Perhaps, it can be hidden using codecvt, but it is rather important to get the optimal size of the Trie. Is it encoded using the fixed-size, 4B, values? Or using the variable-size, 1-4B, values?
   2. Why the number of nodes with 0 children dominate in the case of Czech and Chinese, whereas it is 1 for Vietnamese and English (see page 33)?
   3. Is is not clear why the results of Non-sorted array are always worst although you show that the majority of nodes includes only one value (see page 33).

4. In the case of the Sorted array node, you propose 0.013s for 10 000 queries over the mix data collection, in the case of the Sequential array it is 105.8s. It is not clear why it is 0.011 and 0.036 for prefix searching, i.e. the Sorted array node is only 3x faster?
5. Can you compare your work with the article from the assignment: "Aoe, J.-I., Morimoto, K. and Sato, T. (1992), An efficient implementation of trie structures. Softw: Pract. Exper., 22: 695-721. https://doi.org/10.1002/spe.4380220902"?

Overall classification: excellent

Ostrava, 25.05.2025

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